Filter Summary Report: CG,TIA,Automated,simple,Z1,Z3

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Contents

1 Examined $H(z)$ for CG TIA Automated simple Z1 Z3: $\frac{Z_1 Z_3 g_m}{Z_1 g_m+1}$

$$H(z) = \frac{Z_1 Z_3 g_m}{Z_1 g_m + 1}$$

2 AP

3 BP

3.1 BP-1 $Z(s) = \left(R_1,\ \infty,\ \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{L_3 R_1 R_3 g_m s}{R_1 R_3 g_m + R_3 + s^2 \left(C_3 L_3 R_1 R_3 g_m + C_3 L_3 R_3\right) + s \left(L_3 R_1 g_m + L_3\right)}$$

Parameters:

Q: $\frac{\sqrt{C_3} R_3}{\sqrt{L_3}}$
wo: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$
bandwidth: $\frac{1}{C_3 R_3}$
K-LP: 0
K-HP: 0
K-BP: $\frac{R_1 R_3 g_m}{R_1 g_m + 1}$
Qz: None
Wz: None

3.2 BP-2 $Z(s) = \left(L_1 s,\ \infty,\ \frac{R_3}{C_3 R_3 s + 1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{L_1 R_3 g_m s}{C_3 L_1 R_3 g_m s^2 + s \left(C_3 R_3 + L_1 g_m\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_3}\sqrt{L_1}\sqrt{R_3} g_m \sqrt{\frac{1}{g_m}}}{C_3 R_3 + L_1 g_m}$
wo: $\frac{\sqrt{\frac{1}{g_m}}}{\sqrt{C_3}\sqrt{L_1}\sqrt{R_3}}$
bandwidth: $\frac{C_3 R_3 + L_1 g_m}{C_3 L_1 R_3 g_m}$
K-LP: 0
K-HP: 0
K-BP: $\frac{L_1 R_3 g_m}{C_3 R_3 + L_1 g_m}$
Qz: None
Wz: None

3.3 BP-3 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1},\ \infty,\ R_3,\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{L_1 R_3 g_m s}{C_1 L_1 s^2 + L_1 g_m s + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1}}{\sqrt{L_1 g_m}}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{g_m}{C_1}$
K-LP: 0
K-HP: 0
K-BP: R_3
Qz: None
Wz: None

3.4 BP-4 $Z(s) = \left(\frac{L_1R_1s}{C_1L_1R_1s^2+L_1s+R_1},\ \infty,\ R_3,\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{L_1R_1R_3g_ms}{C_1L_1R_1s^2 + R_1 + s\left(L_1R_1g_m + L_1\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}R_1}{\sqrt{L_1}R_1g_m+\sqrt{L_1}}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{\sqrt{L_1}R_1g_m+\sqrt{L_1}}{C_1\sqrt{L_1}R_1}$
K-LP: 0
K-HP: 0
K-BP: $\frac{R_1R_3g_m}{R_1g_m+1}$
Qz: None
Wz: None

4 BS

4.1 BS-1 $Z(s) = \left(R_1,\ \infty,\ \frac{R_3\left(C_3L_3s^2+1\right)}{C_3L_3s^2+C_3R_3s+1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3L_3R_1R_3g_ms^2 + R_1R_3g_m}{R_1g_m + s^2\left(C_3L_3R_1g_m + C_3L_3\right) + s\left(C_3R_1R_3g_m + C_3R_3\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{L_3}}{\sqrt{C_3}R_3}$
wo: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$
bandwidth: $\frac{R_3}{L_3}$
K-LP: $\frac{R_1R_3g_m}{R_1g_m+1}$
K-HP: $\frac{R_1R_3g_m}{R_1g_m+1}$
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$

4.2 BS-2 $Z(s) = \left(L_1s + \frac{1}{C_1s},\ \infty,\ R_3,\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1L_1R_3g_ms^2 + R_3g_m}{C_1L_1g_ms^2 + C_1s + g_m}$$

Parameters:

Q: $\frac{\sqrt{L_1}g_m}{\sqrt{C_1}}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{1}{L_1g_m}$
K-LP: R_3
K-HP: R_3
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$

4.3 BS-3 $Z(s) = \left(\frac{R_1\left(C_1 L_1 s^2+1\right)}{C_1 L_1 s^2+C_1 R_1 s+1},\ \infty,\ R_3,\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 L_1 R_1 R_3 g_m s^2 + R_1 R_3 g_m}{C_1 R_1 s + R_1 g_m + s^2 \left(C_1 L_1 R_1 g_m + C_1 L_1\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{L_1} R_1 g_m + \sqrt{L_1}}{\sqrt{C_1} R_1}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{R_1}{\sqrt{L_1}\left(\sqrt{L_1} R_1 g_m + \sqrt{L_1}\right)}$
K-LP: $\frac{R_1 R_3 g_m}{R_1 g_m + 1}$
K-HP: $\frac{R_1 R_3 g_m}{R_1 g_m + 1}$
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$

5 GE

5.1 GE-1 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s},\ \infty,\ R_3,\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 L_1 R_3 g_m s^2 + C_1 R_1 R_3 g_m s + R_3 g_m}{C_1 L_1 g_m s^2 + g_m + s\left(C_1 R_1 g_m + C_1\right)}$$

Parameters:

Q: $\frac{\sqrt{L_1} g_m}{\sqrt{C_1} R_1 g_m + \sqrt{C_1}}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{\sqrt{C_1} R_1 g_m + \sqrt{C_1}}{\sqrt{C_1} L_1 g_m}$
K-LP: R_3
K-HP: R_3
K-BP: $\frac{R_1 R_3 g_m}{R_1 g_m + 1}$
Qz: $\frac{\sqrt{L_1}}{\sqrt{C_1} R_1}$
Wz: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$

5.2 GE-2 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1},\ \infty,\ R_3,\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 L_1 R_1 R_3 g_m s^2 + L_1 R_3 g_m s + R_1 R_3 g_m}{L_1 g_m s + R_1 g_m + s^2 \left(C_1 L_1 R_1 g_m + C_1 L_1\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} R_1 g_m + \sqrt{C_1}}{\sqrt{L_1} g_m}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{g_m}{\sqrt{C_1}\left(\sqrt{C_1} R_1 g_m + \sqrt{C_1}\right)}$
K-LP: $\frac{R_1 R_3 g_m}{R_1 g_m + 1}$
K-HP: $\frac{R_1 R_3 g_m}{R_1 g_m + 1}$
K-BP: R_3
Qz: $\frac{\sqrt{C_1} R_1}{\sqrt{L_1}}$
Wz: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$

6 HP

7 LP

7.1 LP-1 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{R_3 g_m}{C_1 C_3 R_3 s^2 + g_m + s\left(C_1 + C_3 R_3 g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_3}\sqrt{R_3}\sqrt{g_m}}{C_1+C_3 R_3 g_m}$
wo: $\frac{\sqrt{g_m}}{\sqrt{C_1}\sqrt{C_3}\sqrt{R_3}}$
bandwidth: $\frac{C_1+C_3 R_3 g_m}{C_1 C_3 R_3}$
K-LP: R_3
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.2 LP-2 $Z(s) = \left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{R_1 R_3 g_m}{C_1 C_3 R_1 R_3 s^2 + R_1 g_m + s\left(C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_3\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_3}\sqrt{R_1}\sqrt{R_3}\sqrt{R_1 g_m+1}}{C_1 R_1+C_3 R_1 R_3 g_m+C_3 R_3}$
wo: $\frac{\sqrt{R_1 g_m+1}}{\sqrt{C_1}\sqrt{C_3}\sqrt{R_1}\sqrt{R_3}}$
bandwidth: $\frac{C_1 R_1+C_3 R_1 R_3 g_m+C_3 R_3}{C_1 C_3 R_1 R_3}$
K-LP: $\frac{R_1 R_3 g_m}{R_1 g_m+1}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.3 LP-3 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2+1},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{L_1 g_m}{C_1 C_3 L_1 s^2 + C_3 L_1 g_m s + C_3}$$

Parameters:

Q: $\frac{\sqrt{C_1}}{\sqrt{L_1} g_m}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{g_m}{C_1}$
K-LP: $\frac{L_1 g_m}{C_3}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.4 LP-4 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2+L_1 s+R_1},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{L_1 R_1 g_m}{C_1 C_3 L_1 R_1 s^2 + C_3 R_1 + s\left(C_3 L_1 R_1 g_m + C_3 L_1\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1} R_1}{\sqrt{L_1} R_1 g_m+\sqrt{L_1}}$

wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{\sqrt{L_1}R_1 g_m+\sqrt{L_1}}{C_1\sqrt{L_1}R_1}$
K-LP: $\frac{L_1 g_m}{C_3}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 R_1 R_3 g_m s + R_3 g_m}{g_m + s^2 \left(C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3\right) + s \left(C_1 R_1 g_m + C_1 + C_3 R_3 g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_3}R_1\sqrt{R_3}g_m\sqrt{\frac{g_m}{R_1 g_m+1}}+\sqrt{C_1}\sqrt{C_3}\sqrt{R_3}\sqrt{\frac{g_m}{R_1 g_m+1}}}{C_1 R_1 g_m+C_1+C_3 R_3 g_m}$
wo: $\sqrt{\frac{g_m}{C_1 C_3 R_1 R_3 g_m+C_1 C_3 R_3}}$
bandwidth: $\frac{\sqrt{\frac{g_m}{C_1 C_3 R_1 R_3 g_m+C_1 C_3 R_3}}(C_1 R_1 g_m+C_1+C_3 R_3 g_m)}{\sqrt{C_1}\sqrt{C_3}R_1\sqrt{R_3}g_m\sqrt{\frac{g_m}{R_1 g_m+1}}+\sqrt{C_1}\sqrt{C_3}\sqrt{R_3}\sqrt{\frac{g_m}{R_1 g_m+1}}}$
K-LP: R_3
K-HP: 0
K-BP: $\frac{C_1 R_1 R_3 g_m}{C_1 R_1 g_m+C_1+C_3 R_3 g_m}$
Qz: None
Wz: None

8.2 X-INVALID-NUMER-2 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2+1},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3 L_1 R_3 g_m s + L_1 g_m}{C_1 C_3 L_1 s^2 + C_3 L_1 g_m s + C_3}$$

Parameters:

Q: $\frac{\sqrt{C_1}}{\sqrt{L_1}g_m}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{g_m}{C_1}$
K-LP: $\frac{L_1 g_m}{C_3}$
K-HP: 0
K-BP: R_3
Qz: None
Wz: None

8.3 X-INVALID-NUMER-3 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2+L_1 s+R_1},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3 L_1 R_1 R_3 g_m s + L_1 R_1 g_m}{C_1 C_3 L_1 R_1 s^2 + C_3 R_1 + s \left(C_3 L_1 R_1 g_m + C_3 L_1\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}R_1}{\sqrt{L_1}R_1 g_m+\sqrt{L_1}}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{\sqrt{L_1}R_1 g_m+\sqrt{L_1}}{C_1\sqrt{L_1}R_1}$
K-LP: $\frac{L_1 g_m}{C_3}$
K-HP: 0
K-BP: $\frac{R_1 R_3 g_m}{R_1 g_m+1}$

Qz: None
Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (R_1,\ \infty,\ R_3,\ \infty,\ \infty,\ \infty)$

$$H(s) = \frac{R_1 R_3 g_m}{R_1 g_m + 1}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(R_1,\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{R_1 g_m}{s\left(C_3 R_1 g_m + C_3\right)}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(R_1,\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{R_1 R_3 g_m}{R_1 g_m + s\left(C_3 R_1 R_3 g_m + C_3 R_3\right) + 1}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(R_1,\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3 R_1 R_3 g_m s + R_1 g_m}{s\left(C_3 R_1 g_m + C_3\right)}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(R_1,\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3 L_3 R_1 g_m s^2 + R_1 g_m}{s\left(C_3 R_1 g_m + C_3\right)}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(R_1,\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2+1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{L_3 R_1 g_m s}{R_1 g_m + s^2\left(C_3 L_3 R_1 g_m + C_3 L_3\right) + 1}$$

9.7 X-INVALID-ORDER-7 $Z(s) = \left(R_1,\ \infty,\ L_3 s + R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3 L_3 R_1 g_m s^2 + C_3 R_1 R_3 g_m s + R_1 g_m}{s\left(C_3 R_1 g_m + C_3\right)}$$

9.8 X-INVALID-ORDER-8 $Z(s) = \left(R_1,\ \infty,\ \frac{C_3 L_3 R_3 s^2+L_3 s+R_3}{C_3 L_3 s^2+1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3 L_3 R_1 R_3 g_m s^2 + L_3 R_1 g_m s + R_1 R_3 g_m}{R_1 g_m + s^2\left(C_3 L_3 R_1 g_m + C_3 L_3\right) + 1}$$

9.9 X-INVALID-ORDER-9 $Z(s) = (L_1 s,\ \infty,\ R_3,\ \infty,\ \infty,\ \infty)$

$$H(s) = \frac{L_1 R_3 g_m s}{L_1 g_m s + 1}$$

9.10 X-INVALID-ORDER-10 $Z(s) = \left(L_1 s,\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{L_1 g_m}{C_3 L_1 g_m s + C_3}$$

9.11 X-INVALID-ORDER-11 $Z(s) = \left(L_1 s,\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3 L_1 R_3 g_m s + L_1 g_m}{C_3 L_1 g_m s + C_3}$$

9.12 X-INVALID-ORDER-12 $Z(s) = \left(L_1 s,\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3 L_1 L_3 g_m s^2 + L_1 g_m}{C_3 L_1 g_m s + C_3}$$

9.13 X-INVALID-ORDER-13 $Z(s) = \left(L_1 s,\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2 + 1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{L_1 L_3 g_m s^2}{C_3 L_1 L_3 g_m s^3 + C_3 L_3 s^2 + L_1 g_m s + 1}$$

9.14 X-INVALID-ORDER-14 $Z(s) = \left(L_1 s,\ \infty,\ L_3 s + R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3 L_1 L_3 g_m s^2 + C_3 L_1 R_3 g_m s + L_1 g_m}{C_3 L_1 g_m s + C_3}$$

9.15 X-INVALID-ORDER-15 $Z(s) = \left(L_1 s,\ \infty,\ \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{L_1 L_3 R_3 g_m s^2}{C_3 L_1 L_3 R_3 g_m s^3 + R_3 + s^2 \left(C_3 L_3 R_3 + L_1 L_3 g_m\right) + s \left(L_1 R_3 g_m + L_3\right)}$$

9.16 X-INVALID-ORDER-16 $Z(s) = \left(L_1 s,\ \infty,\ \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3 L_1 L_3 R_3 g_m s^3 + L_1 L_3 g_m s^2 + L_1 R_3 g_m s}{C_3 L_1 L_3 g_m s^3 + C_3 L_3 s^2 + L_1 g_m s + 1}$$

9.17 X-INVALID-ORDER-17 $Z(s) = \left(L_1 s,\ \infty,\ \frac{R_3 \left(C_3 L_3 s^2 + 1\right)}{C_3 L_3 s^2 + C_3 R_3 s + 1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3 L_1 L_3 R_3 g_m s^3 + L_1 R_3 g_m s}{C_3 L_1 L_3 g_m s^3 + s^2 \left(C_3 L_1 R_3 g_m + C_3 L_3\right) + s \left(C_3 R_3 + L_1 g_m\right) + 1}$$

9.18 X-INVALID-ORDER-18 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ R_3,\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{R_3 g_m}{C_1 s + g_m}$$

9.19 X-INVALID-ORDER-19 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{g_m}{C_1 C_3 s^2 + C_3 g_m s}$$

9.20 X-INVALID-ORDER-20 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3 R_3 g_m s + g_m}{C_1 C_3 s^2 + C_3 g_m s}$$

9.21 X-INVALID-ORDER-21 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3 L_3 g_m s^2 + g_m}{C_1 C_3 s^2 + C_3 g_m s}$$

9.22 X-INVALID-ORDER-22 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2 + 1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{L_3 g_m s}{C_1 C_3 L_3 s^3 + C_1 s + C_3 L_3 g_m s^2 + g_m}$$

9.23 X-INVALID-ORDER-23 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ L_3 s + R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3 L_3 g_m s^2 + C_3 R_3 g_m s + g_m}{C_1 C_3 s^2 + C_3 g_m s}$$

9.24 X-INVALID-ORDER-24 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{L_3 R_3 g_m s}{C_1 C_3 L_3 R_3 s^3 + R_3 g_m + s^2 \left(C_1 L_3 + C_3 L_3 R_3 g_m\right) + s \left(C_1 R_3 + L_3 g_m\right)}$$

9.25 X-INVALID-ORDER-25 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3 L_3 R_3 g_m s^2 + L_3 g_m s + R_3 g_m}{C_1 C_3 L_3 s^3 + C_1 s + C_3 L_3 g_m s^2 + g_m}$$

9.26 X-INVALID-ORDER-26 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \frac{R_3 \left(C_3 L_3 s^2 + 1\right)}{C_3 L_3 s^2 + C_3 R_3 s + 1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3 L_3 R_3 g_m s^2 + R_3 g_m}{C_1 C_3 L_3 s^3 + g_m + s^2 \left(C_1 C_3 R_3 + C_3 L_3 g_m\right) + s \left(C_1 + C_3 R_3 g_m\right)}$$

9.27 X-INVALID-ORDER-27 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1},\ \infty,\ R_3,\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{R_1 R_3 g_m}{C_1 R_1 s + R_1 g_m + 1}$$

9.28 X-INVALID-ORDER-28 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{R_1 g_m}{C_1 C_3 R_1 s^2 + s \left(C_3 R_1 g_m + C_3\right)}$$

9.29 X-INVALID-ORDER-29 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ R_3 + \frac{1}{C_3s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3R_1R_3g_ms + R_1g_m}{C_1C_3R_1s^2 + s\left(C_3R_1g_m + C_3\right)}$$

9.30 X-INVALID-ORDER-30 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ L_3s + \frac{1}{C_3s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3L_3R_1g_ms^2 + R_1g_m}{C_1C_3R_1s^2 + s\left(C_3R_1g_m + C_3\right)}$$

9.31 X-INVALID-ORDER-31 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ \frac{L_3s}{C_3L_3s^2+1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{L_3R_1g_ms}{C_1C_3L_3R_1s^3 + C_1R_1s + R_1g_m + s^2\left(C_3L_3R_1g_m + C_3L_3\right) + 1}$$

9.32 X-INVALID-ORDER-32 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ L_3s + R_3 + \frac{1}{C_3s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3L_3R_1g_ms^2 + C_3R_1R_3g_ms + R_1g_m}{C_1C_3R_1s^2 + s\left(C_3R_1g_m + C_3\right)}$$

9.33 X-INVALID-ORDER-33 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ \frac{L_3R_3s}{C_3L_3R_3s^2+L_3s+R_3},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{L_3R_1R_3g_ms}{C_1C_3L_3R_1R_3s^3 + R_1R_3g_m + R_3 + s^2\left(C_1L_3R_1 + C_3L_3R_1R_3g_m + C_3L_3R_3\right) + s\left(C_1R_1R_3 + L_3R_1g_m + L_3\right)}$$

9.34 X-INVALID-ORDER-34 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ \frac{C_3L_3R_3s^2+L_3s+R_3}{C_3L_3s^2+1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3L_3R_1R_3g_ms^2 + L_3R_1g_ms + R_1R_3g_m}{C_1C_3L_3R_1s^3 + C_1R_1s + R_1g_m + s^2\left(C_3L_3R_1g_m + C_3L_3\right) + 1}$$

9.35 X-INVALID-ORDER-35 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ \frac{R_3\left(C_3L_3s^2+1\right)}{C_3L_3s^2+C_3R_3s+1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3L_3R_1R_3g_ms^2 + R_1R_3g_m}{C_1C_3L_3R_1s^3 + R_1g_m + s^2\left(C_1C_3R_1R_3 + C_3L_3R_1g_m + C_3L_3\right) + s\left(C_1R_1 + C_3R_1R_3g_m + C_3R_3\right) + 1}$$

9.36 X-INVALID-ORDER-36 $Z(s) = \left(R_1 + \frac{1}{C_1s},\ \infty,\ R_3,\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1R_1R_3g_ms + R_3g_m}{g_m + s\left(C_1R_1g_m + C_1\right)}$$

9.37 X-INVALID-ORDER-37 $Z(s) = \left(R_1 + \frac{1}{C_1s},\ \infty,\ \frac{1}{C_3s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1R_1g_ms + g_m}{C_3g_ms + s^2\left(C_1C_3R_1g_m + C_1C_3\right)}$$

9.38 X-INVALID-ORDER-38 $Z(s) = \left(R_1 + \frac{1}{C_1s},\ \infty,\ R_3 + \frac{1}{C_3s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1C_3R_1R_3g_ms^2 + g_m + s\left(C_1R_1g_m + C_3R_3g_m\right)}{C_3g_ms + s^2\left(C_1C_3R_1g_m + C_1C_3\right)}$$

9.39 X-INVALID-ORDER-39 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 C_3 L_3 R_1 g_m s^3 + C_1 R_1 g_m s + C_3 L_3 g_m s^2 + g_m}{C_3 g_m s + s^2 \left(C_1 C_3 R_1 g_m + C_1 C_3\right)}$$

9.40 X-INVALID-ORDER-40 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2 + 1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 L_3 R_1 g_m s^2 + L_3 g_m s}{C_3 L_3 g_m s^2 + g_m + s^3 \left(C_1 C_3 L_3 R_1 g_m + C_1 C_3 L_3\right) + s \left(C_1 R_1 g_m + C_1\right)}$$

9.41 X-INVALID-ORDER-41 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ L_3 s + R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 C_3 L_3 R_1 g_m s^3 + g_m + s^2 \left(C_1 C_3 R_1 R_3 g_m + C_3 L_3 g_m\right) + s \left(C_1 R_1 g_m + C_3 R_3 g_m\right)}{C_3 g_m s + s^2 \left(C_1 C_3 R_1 g_m + C_1 C_3\right)}$$

9.42 X-INVALID-ORDER-42 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 L_3 R_1 R_3 g_m s^2 + L_3 R_3 g_m s}{R_3 g_m + s^3 \left(C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_3 L_3 R_3\right) + s^2 \left(C_1 L_3 R_1 g_m + C_1 L_3 + C_3 L_3 R_3 g_m\right) + s \left(C_1 R_1 R_3 g_m + C_1 R_3 + L_3 g_m\right)}$$

9.43 X-INVALID-ORDER-43 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 C_3 L_3 R_1 R_3 g_m s^3 + R_3 g_m + s^2 \left(C_1 L_3 R_1 g_m + C_3 L_3 R_3 g_m\right) + s \left(C_1 R_1 R_3 g_m + L_3 g_m\right)}{C_3 L_3 g_m s^2 + g_m + s^3 \left(C_1 C_3 L_3 R_1 g_m + C_1 C_3 L_3\right) + s \left(C_1 R_1 g_m + C_1\right)}$$

9.44 X-INVALID-ORDER-44 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{R_3 \left(C_3 L_3 s^2 + 1\right)}{C_3 L_3 s^2 + C_3 R_3 s + 1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 C_3 L_3 R_1 R_3 g_m s^3 + C_1 R_1 R_3 g_m s + C_3 L_3 R_3 g_m s^2 + R_3 g_m}{g_m + s^3 \left(C_1 C_3 L_3 R_1 g_m + C_1 C_3 L_3\right) + s^2 \left(C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3 + C_3 L_3 g_m\right) + s \left(C_1 R_1 g_m + C_1 + C_3 R_3 g_m\right)}$$

9.45 X-INVALID-ORDER-45 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 L_1 g_m s^2 + g_m}{C_1 C_3 L_1 g_m s^3 + C_1 C_3 s^2 + C_3 g_m s}$$

9.46 X-INVALID-ORDER-46 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \frac{R_3}{C_3 R_3 s + 1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 L_1 R_3 g_m s^2 + R_3 g_m}{C_1 C_3 L_1 R_3 g_m s^3 + g_m + s^2 \left(C_1 C_3 R_3 + C_1 L_1 g_m\right) + s \left(C_1 + C_3 R_3 g_m\right)}$$

9.47 X-INVALID-ORDER-47 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 C_3 L_1 R_3 g_m s^3 + C_1 L_1 g_m s^2 + C_3 R_3 g_m s + g_m}{C_1 C_3 L_1 g_m s^3 + C_1 C_3 s^2 + C_3 g_m s}$$

9.48 X-INVALID-ORDER-48 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 g_m s^4 + g_m + s^2 \left(C_1 L_1 g_m + C_3 L_3 g_m\right)}{C_1 C_3 L_1 g_m s^3 + C_1 C_3 s^2 + C_3 g_m s}$$

9.49 X-INVALID-ORDER-49 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2 + 1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 L_1 L_3 g_m s^3 + L_3 g_m s}{C_1 C_3 L_1 L_3 g_m s^4 + C_1 C_3 L_3 s^3 + C_1 s + g_m + s^2 \left(C_1 L_1 g_m + C_3 L_3 g_m\right)}$$

9.50 X-INVALID-ORDER-50 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ L_3 s + R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 g_m s^4 + C_1 C_3 L_1 R_3 g_m s^3 + C_3 R_3 g_m s + g_m + s^2 \left(C_1 L_1 g_m + C_3 L_3 g_m\right)}{C_1 C_3 L_1 g_m s^3 + C_1 C_3 s^2 + C_3 g_m s}$$

9.51 X-INVALID-ORDER-51 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 L_1 L_3 R_3 g_m s^3 + L_3 R_3 g_m s}{C_1 C_3 L_1 L_3 R_3 g_m s^4 + R_3 g_m + s^3 \left(C_1 C_3 L_3 R_3 + C_1 L_1 L_3 g_m\right) + s^2 \left(C_1 L_1 R_3 g_m + C_1 L_3 + C_3 L_3 R_3 g_m\right) + s \left(C_1 R_3 + L_3 g_m\right)}$$

9.52 X-INVALID-ORDER-52 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_3 g_m s^4 + C_1 L_1 L_3 g_m s^3 + L_3 g_m s + R_3 g_m + s^2 \left(C_1 L_1 R_3 g_m + C_3 L_3 R_3 g_m\right)}{C_1 C_3 L_1 L_3 g_m s^4 + C_1 C_3 L_3 s^3 + C_1 s + g_m + s^2 \left(C_1 L_1 g_m + C_3 L_3 g_m\right)}$$

9.53 X-INVALID-ORDER-53 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \frac{R_3 \left(C_3 L_3 s^2 + 1\right)}{C_3 L_3 s^2 + C_3 R_3 s + 1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_3 g_m s^4 + R_3 g_m + s^2 \left(C_1 L_1 R_3 g_m + C_3 L_3 R_3 g_m\right)}{C_1 C_3 L_1 L_3 g_m s^4 + g_m + s^3 \left(C_1 C_3 L_1 R_3 g_m + C_1 C_3 L_3\right) + s^2 \left(C_1 C_3 R_3 + C_1 L_1 g_m + C_3 L_3 g_m\right) + s \left(C_1 + C_3 R_3 g_m\right)}$$

9.54 X-INVALID-ORDER-54 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1},\ \infty,\ \frac{R_3}{C_3 R_3 s + 1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{L_1 R_3 g_m s}{C_1 C_3 L_1 R_3 s^3 + s^2 \left(C_1 L_1 + C_3 L_1 R_3 g_m\right) + s \left(C_3 R_3 + L_1 g_m\right) + 1}$$

9.55 X-INVALID-ORDER-55 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2 + 1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{L_1 L_3 g_m s^2}{C_1 C_3 L_1 L_3 s^4 + C_3 L_1 L_3 g_m s^3 + L_1 g_m s + s^2 \left(C_1 L_1 + C_3 L_3\right) + 1}$$

9.56 X-INVALID-ORDER-56 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1},\ \infty,\ \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{L_1 L_3 R_3 g_m s^2}{C_1 C_3 L_1 L_3 R_3 s^4 + R_3 + s^3 \left(C_1 L_1 L_3 + C_3 L_1 L_3 R_3 g_m\right) + s^2 \left(C_1 L_1 R_3 + C_3 L_3 R_3 + L_1 L_3 g_m\right) + s \left(L_1 R_3 g_m + L_3\right)}$$

9.57 X-INVALID-ORDER-57 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1},\ \infty,\ \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3 L_1 L_3 R_3 g_m s^3 + L_1 L_3 g_m s^2 + L_1 R_3 g_m s}{C_1 C_3 L_1 L_3 s^4 + C_3 L_1 L_3 g_m s^3 + L_1 g_m s + s^2 \left(C_1 L_1 + C_3 L_3\right) + 1}$$

9.58 X-INVALID-ORDER-58 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2+1},\ \infty,\ \frac{R_3\left(C_3 L_3 s^2+1\right)}{C_3 L_3 s^2+C_3 R_3 s+1},\ \infty,\ \infty,\ \infty \right)$

$$H(s) = \frac{C_3 L_1 L_3 R_3 g_m s^3 + L_1 R_3 g_m s}{C_1 C_3 L_1 L_3 s^4 + s^3 \left(C_1 C_3 L_1 R_3 + C_3 L_1 L_3 g_m\right) + s^2 \left(C_1 L_1 + C_3 L_1 R_3 g_m + C_3 L_3\right) + s \left(C_3 R_3 + L_1 g_m\right) + 1}$$

9.59 X-INVALID-ORDER-59 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty \right)$

$$H(s) = \frac{C_1 L_1 g_m s^2 + C_1 R_1 g_m s + g_m}{C_1 C_3 L_1 g_m s^3 + C_3 g_m s + s^2 \left(C_1 C_3 R_1 g_m + C_1 C_3\right)}$$

9.60 X-INVALID-ORDER-60 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ \infty \right)$

$$H(s) = \frac{C_1 L_1 R_3 g_m s^2 + C_1 R_1 R_3 g_m s + R_3 g_m}{C_1 C_3 L_1 R_3 g_m s^3 + g_m + s^2 \left(C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3 + C_1 L_1 g_m\right) + s \left(C_1 R_1 g_m + C_1 + C_3 R_3 g_m\right)}$$

9.61 X-INVALID-ORDER-61 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 R_3 g_m s^3 + g_m + s^2 \left(C_1 C_3 R_1 R_3 g_m + C_1 L_1 g_m\right) + s \left(C_1 R_1 g_m + C_3 R_3 g_m\right)}{C_1 C_3 L_1 g_m s^3 + C_3 g_m s + s^2 \left(C_1 C_3 R_1 g_m + C_1 C_3\right)}$$

9.62 X-INVALID-ORDER-62 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 g_m s^4 + C_1 C_3 L_3 R_1 g_m s^3 + C_1 R_1 g_m s + g_m + s^2 \left(C_1 L_1 g_m + C_3 L_3 g_m\right)}{C_1 C_3 L_1 g_m s^3 + C_3 g_m s + s^2 \left(C_1 C_3 R_1 g_m + C_1 C_3\right)}$$

9.63 X-INVALID-ORDER-63 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2+1},\ \infty,\ \infty,\ \infty \right)$

$$H(s) = \frac{C_1 L_1 L_3 g_m s^3 + C_1 L_3 R_1 g_m s^2 + L_3 g_m s}{C_1 C_3 L_1 L_3 g_m s^4 + g_m + s^3 \left(C_1 C_3 L_3 R_1 g_m + C_1 C_3 L_3\right) + s^2 \left(C_1 L_1 g_m + C_3 L_3 g_m\right) + s \left(C_1 R_1 g_m + C_1\right)}$$

9.64 X-INVALID-ORDER-64 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s},\ \infty,\ L_3 s + R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 g_m s^4 + g_m + s^3 \left(C_1 C_3 L_1 R_3 g_m + C_1 C_3 L_3 R_1 g_m\right) + s^2 \left(C_1 C_3 R_1 R_3 g_m + C_1 L_1 g_m + C_3 L_3 g_m\right) + s \left(C_1 R_1 g_m + C_3 R_3 g_m\right)}{C_1 C_3 L_1 g_m s^3 + C_3 g_m s + s^2 \left(C_1 C_3 R_1 g_m + C_1 C_3\right)}$$

9.65 X-INVALID-ORDER-65 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2+L_3 s+R_3},\ \infty,\ \infty,\ \infty \right)$

$$H(s) = \frac{C_1 L_1 L_3 R_3 g_m s^3 + C_1 L_3 R_1 R_3 g_m s^2 + L_3 R_3 g_m s}{C_1 C_3 L_1 L_3 R_3 g_m s^4 + R_3 g_m + s^3 \left(C_1 C_3 L_3 R_1 R_3 g_m + C_1 C_3 L_3 R_3 + C_1 L_1 L_3 g_m\right) + s^2 \left(C_1 L_1 R_3 g_m + C_1 L_3 R_1 g_m + C_1 L_3 + C_3 L_3 R_3 g_m\right) + s \left(C_1 R_1 R_3 g_m + C_1 R_3 + L_3 g_m\right)}$$

9.66 X-INVALID-ORDER-66 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{C_3 L_3 R_3 s^2+L_3 s+R_3}{C_3 L_3 s^2+1},\ \infty,\ \infty,\ \infty \right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_3 g_m s^4 + R_3 g_m + s^3 \left(C_1 C_3 L_3 R_1 R_3 g_m + C_1 L_1 L_3 g_m\right) + s^2 \left(C_1 L_1 R_3 g_m + C_1 L_3 R_1 g_m + C_3 L_3 R_3 g_m\right) + s \left(C_1 R_1 R_3 g_m + L_3 g_m\right)}{C_1 C_3 L_1 L_3 g_m s^4 + g_m + s^3 \left(C_1 C_3 L_3 R_1 g_m + C_1 C_3 L_3\right) + s^2 \left(C_1 L_1 g_m + C_3 L_3 g_m\right) + s \left(C_1 R_1 g_m + C_1\right)}$$

9.67 X-INVALID-ORDER-67 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{R_3\left(C_3 L_3 s^2+1\right)}{C_3 L_3 s^2+C_3 R_3 s+1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 C_3 L_1 L_3 R_3 g_m s^4 + C_1 C_3 L_3 R_1 R_3 g_m s^3 + C_1 R_1 R_3 g_m s + R_3 g_m + s^2\left(C_1 L_1 R_3 g_m + C_3 L_3 R_3 g_m\right)}{C_1 C_3 L_1 L_3 g_m s^4 + g_m + s^3\left(C_1 C_3 L_1 R_3 g_m + C_1 C_3 L_3 R_1 g_m + C_1 C_3 L_3\right) + s^2\left(C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3 + C_1 L_1 g_m + C_3 L_3 g_m\right) + s\left(C_1 R_1 g_m + C_1 + C_3 R_3 g_m\right)}$$

9.68 X-INVALID-ORDER-68 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2+L_1 s+R_1},\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{L_1 R_1 R_3 g_m s}{C_1 C_3 L_1 R_1 R_3 s^3 + R_1 + s^2\left(C_1 L_1 R_1 + C_3 L_1 R_1 R_3 g_m + C_3 L_1 R_3\right) + s\left(C_3 R_1 R_3 + L_1 R_1 g_m + L_1\right)}$$

9.69 X-INVALID-ORDER-69 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2+L_1 s+R_1},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2+1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{L_1 L_3 R_1 g_m s^2}{C_1 C_3 L_1 L_3 R_1 s^4 + R_1 + s^3\left(C_3 L_1 L_3 R_1 g_m + C_3 L_1 L_3\right) + s^2\left(C_1 L_1 R_1 + C_3 L_3 R_1\right) + s\left(L_1 R_1 g_m + L_1\right)}$$

9.70 X-INVALID-ORDER-70 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2+L_1 s+R_1},\ \infty,\ \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2+L_3 s+R_3},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{L_1 L_3 R_1 R_3 g_m s^2}{C_1 C_3 L_1 L_3 R_1 R_3 s^4 + R_1 R_3 + s^3\left(C_1 L_1 L_3 R_1 + C_3 L_1 L_3 R_1 R_3 g_m + C_3 L_1 L_3 R_3\right) + s^2\left(C_1 L_1 R_1 R_3 + C_3 L_3 R_1 R_3 + L_1 L_3 R_1 g_m + L_1 L_3\right) + s\left(L_1 R_1 R_3 g_m + L_1 R_3 + L_3 R_1\right)}$$

9.71 X-INVALID-ORDER-71 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2+L_1 s+R_1},\ \infty,\ \frac{C_3 L_3 R_3 s^2+L_3 s+R_3}{C_3 L_3 s^2+1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3 L_1 L_3 R_1 R_3 g_m s^3 + L_1 L_3 R_1 g_m s^2 + L_1 R_1 R_3 g_m s}{C_1 C_3 L_1 L_3 R_1 s^4 + R_1 + s^3\left(C_3 L_1 L_3 R_1 g_m + C_3 L_1 L_3\right) + s^2\left(C_1 L_1 R_1 + C_3 L_3 R_1\right) + s\left(L_1 R_1 g_m + L_1\right)}$$

9.72 X-INVALID-ORDER-72 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2+L_1 s+R_1},\ \infty,\ \frac{R_3\left(C_3 L_3 s^2+1\right)}{C_3 L_3 s^2+C_3 R_3 s+1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3 L_1 L_3 R_1 R_3 g_m s^3 + L_1 R_1 R_3 g_m s}{C_1 C_3 L_1 L_3 R_1 s^4 + R_1 + s^3\left(C_1 C_3 L_1 R_1 R_3 + C_3 L_1 L_3 R_1 g_m + C_3 L_1 L_3\right) + s^2\left(C_1 L_1 R_1 + C_3 L_1 R_1 R_3 g_m + C_3 L_1 R_3 + C_3 L_3 R_1\right) + s\left(C_3 R_1 R_3 + L_1 R_1 g_m + L_1\right)}$$

9.73 X-INVALID-ORDER-73 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2+L_1 s+R_1}{C_1 L_1 s^2+1},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 L_1 R_1 g_m s^2 + L_1 g_m s + R_1 g_m}{C_3 L_1 g_m s^2 + s^3\left(C_1 C_3 L_1 R_1 g_m + C_1 C_3 L_1\right) + s\left(C_3 R_1 g_m + C_3\right)}$$

9.74 X-INVALID-ORDER-74 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2+L_1 s+R_1}{C_1 L_1 s^2+1},\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 L_1 R_1 R_3 g_m s^2 + L_1 R_3 g_m s + R_1 R_3 g_m}{R_1 g_m + s^3\left(C_1 C_3 L_1 R_1 R_3 g_m + C_1 C_3 L_1 R_3\right) + s^2\left(C_1 L_1 R_1 g_m + C_1 L_1 + C_3 L_1 R_3 g_m\right) + s\left(C_3 R_1 R_3 g_m + C_3 R_3 + L_1 g_m\right) + 1}$$

9.75 X-INVALID-ORDER-75 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2+L_1 s+R_1}{C_1 L_1 s^2+1},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 C_3 L_1 R_1 R_3 g_m s^3 + R_1 g_m + s^2\left(C_1 L_1 R_1 g_m + C_3 L_1 R_3 g_m\right) + s\left(C_3 R_1 R_3 g_m + L_1 g_m\right)}{C_3 L_1 g_m s^2 + s^3\left(C_1 C_3 L_1 R_1 g_m + C_1 C_3 L_1\right) + s\left(C_3 R_1 g_m + C_3\right)}$$

9.76 X-INVALID-ORDER-76 $Z(s) = \left(\frac{C_1L_1R_1s^2+L_1s+R_1}{C_1L_1s^2+1},\ \infty,\ L_3s+\frac{1}{C_3s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1C_3L_1L_3R_1g_ms^4 + C_3L_1L_3g_ms^3 + L_1g_ms + R_1g_m + s^2\left(C_1L_1R_1g_m + C_3L_3R_1g_m\right)}{C_3L_1g_ms^2 + s^3\left(C_1C_3L_1R_1g_m + C_1C_3L_1\right) + s\left(C_3R_1g_m + C_3\right)}$$

9.77 X-INVALID-ORDER-77 $Z(s) = \left(\frac{C_1L_1R_1s^2+L_1s+R_1}{C_1L_1s^2+1},\ \infty,\ \frac{L_3s}{C_3L_3s^2+1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1L_1L_3R_1g_ms^3 + L_1L_3g_ms^2 + L_3R_1g_ms}{C_3L_1L_3g_ms^3 + L_1g_ms + R_1g_m + s^4\left(C_1C_3L_1L_3R_1g_m + C_1C_3L_1L_3\right) + s^2\left(C_1L_1R_1g_m + C_1L_1 + C_3L_3R_1g_m + C_3L_3\right) + 1}$$

9.78 X-INVALID-ORDER-78 $Z(s) = \left(\frac{C_1L_1R_1s^2+L_1s+R_1}{C_1L_1s^2+1},\ \infty,\ L_3s+R_3+\frac{1}{C_3s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1C_3L_1L_3R_1g_ms^4 + R_1g_m + s^3\left(C_1C_3L_1R_1R_3g_m + C_3L_1L_3g_m\right) + s^2\left(C_1L_1R_1g_m + C_3L_1R_3g_m + C_3L_3R_1g_m\right) + s\left(C_3R_1R_3g_m + L_1g_m\right)}{C_3L_1g_ms^2 + s^3\left(C_1C_3L_1R_1g_m + C_1C_3L_1\right) + s\left(C_3R_1g_m + C_3\right)}$$

9.79 X-INVALID-ORDER-79 $Z(s) = \left(\frac{C_1L_1R_1s^2+L_1s+R_1}{C_1L_1s^2+1},\ \infty,\ \frac{L_3R_3s}{C_3L_3R_3s^2+L_3s+R_3},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1L_1L_3R_1R_3g_ms^3 + L_1L_3R_3g_ms^2 + L_3R_1R_3g_ms}{R_1R_3g_m + R_3 + s^4\left(C_1C_3L_1L_3R_1R_3g_m + C_1C_3L_1L_3R_3\right) + s^3\left(C_1L_1L_3R_1g_m + C_1L_1L_3 + C_3L_1L_3R_3g_m\right) + s^2\left(C_1L_1R_1R_3g_m + C_1L_1R_3 + C_3L_3R_1R_3g_m + C_3L_3R_3 + L_1L_3g_m\right) + s\left(L_1R_3g_m + L_3R_1g_m + L_3\right)}$$

9.80 X-INVALID-ORDER-80 $Z(s) = \left(\frac{C_1L_1R_1s^2+L_1s+R_1}{C_1L_1s^2+1},\ \infty,\ \frac{C_3L_3R_3s^2+L_3s+R_3}{C_3L_3s^2+1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1C_3L_1L_3R_1R_3g_ms^4 + R_1R_3g_m + s^3\left(C_1L_1L_3R_1g_m + C_3L_1L_3R_3g_m\right) + s^2\left(C_1L_1R_1R_3g_m + C_3L_3R_1R_3g_m + L_1L_3g_m\right) + s\left(L_1R_3g_m + L_3R_1g_m\right)}{C_3L_1L_3g_ms^3 + L_1g_ms + R_1g_m + s^4\left(C_1C_3L_1L_3R_1g_m + C_1C_3L_1L_3\right) + s^2\left(C_1L_1R_1g_m + C_1L_1 + C_3L_3R_1g_m + C_3L_3\right) + 1}$$

9.81 X-INVALID-ORDER-81 $Z(s) = \left(\frac{C_1L_1R_1s^2+L_1s+R_1}{C_1L_1s^2+1},\ \infty,\ \frac{R_3\left(C_3L_3s^2+1\right)}{C_3L_3s^2+C_3R_3s+1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1C_3L_1L_3R_1R_3g_ms^4 + C_3L_1L_3R_3g_ms^3 + L_1R_3g_ms + R_1R_3g_m + s^2\left(C_1L_1R_1R_3g_m + C_3L_3R_1R_3g_m\right)}{R_1g_m + s^4\left(C_1C_3L_1L_3R_1g_m + C_1C_3L_1L_3\right) + s^3\left(C_1C_3L_1R_1R_3g_m + C_1C_3L_1R_3 + C_3L_1L_3g_m\right) + s^2\left(C_1L_1R_1g_m + C_1L_1 + C_3L_1R_3g_m + C_3L_3R_1g_m + C_3L_3\right) + s\left(C_3R_1R_3g_m + C_3R_3 + L_1g_m\right) + 1}$$

9.82 X-INVALID-ORDER-82 $Z(s) = \left(\frac{R_1\left(C_1L_1s^2+1\right)}{C_1L_1s^2+C_1R_1s+1},\ \infty,\ \frac{1}{C_3s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1L_1R_1g_ms^2 + R_1g_m}{C_1C_3R_1s^2 + s^3\left(C_1C_3L_1R_1g_m + C_1C_3L_1\right) + s\left(C_3R_1g_m + C_3\right)}$$

9.83 X-INVALID-ORDER-83 $Z(s) = \left(\frac{R_1\left(C_1L_1s^2+1\right)}{C_1L_1s^2+C_1R_1s+1},\ \infty,\ \frac{R_3}{C_3R_3s+1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1L_1R_1R_3g_ms^2 + R_1R_3g_m}{R_1g_m + s^3\left(C_1C_3L_1R_1R_3g_m + C_1C_3L_1R_3\right) + s^2\left(C_1C_3R_1R_3 + C_1L_1R_1g_m + C_1L_1\right) + s\left(C_1R_1 + C_3R_1R_3g_m + C_3R_3\right) + 1}$$

9.84 X-INVALID-ORDER-84 $Z(s) = \left(\frac{R_1\left(C_1L_1s^2+1\right)}{C_1L_1s^2+C_1R_1s+1},\ \infty,\ R_3+\frac{1}{C_3s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1C_3L_1R_1R_3g_ms^3 + C_1L_1R_1g_ms^2 + C_3R_1R_3g_ms + R_1g_m}{C_1C_3R_1s^2 + s^3\left(C_1C_3L_1R_1g_m + C_1C_3L_1\right) + s\left(C_3R_1g_m + C_3\right)}$$

9.85 X-INVALID-ORDER-85 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1},\ \infty,\ L_3s + \frac{1}{C_3s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1C_3L_1L_3R_1g_ms^4 + R_1g_m + s^2(C_1L_1R_1g_m + C_3L_3R_1g_m)}{C_1C_3R_1s^2 + s^3(C_1C_3L_1R_1g_m + C_1C_3L_1) + s(C_3R_1g_m + C_3)}$$

9.86 X-INVALID-ORDER-86 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1},\ \infty,\ \frac{L_3s}{C_3L_3s^2+1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1L_1L_3R_1g_ms^3 + L_3R_1g_ms}{C_1C_3L_3R_1s^3 + C_1R_1s + R_1g_m + s^4(C_1C_3L_1L_3R_1g_m + C_1C_3L_1L_3) + s^2(C_1L_1R_1g_m + C_1L_1 + C_3L_3R_1g_m + C_3L_3) + 1}$$

9.87 X-INVALID-ORDER-87 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1},\ \infty,\ L_3s + R_3 + \frac{1}{C_3s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1C_3L_1L_3R_1g_ms^4 + C_1C_3L_1R_1R_3g_ms^3 + C_3R_1R_3g_ms + R_1g_m + s^2(C_1L_1R_1g_m + C_3L_3R_1g_m)}{C_1C_3R_1s^2 + s^3(C_1C_3L_1R_1g_m + C_1C_3L_1) + s(C_3R_1g_m + C_3)}$$

9.88 X-INVALID-ORDER-88 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1},\ \infty,\ \frac{L_3R_3s}{C_3L_3R_3s^2+L_3s+R_3},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1L_1L_3R_1R_3g_ms^3 + L_3R_1R_3g_ms}{R_1R_3g_m + R_3 + s^4(C_1C_3L_1L_3R_1R_3g_m + C_1C_3L_1L_3R_3) + s^3(C_1C_3L_3R_1R_3 + C_1L_1L_3R_1g_m + C_1L_1L_3) + s^2(C_1L_1R_1R_3g_m + C_1L_1R_3 + C_1L_3R_1 + C_3L_3R_1R_3g_m + C_3L_3R_3) + s(C_1R_1R_3 + L_3R_1g_m + L_3)}$$

9.89 X-INVALID-ORDER-89 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1},\ \infty,\ \frac{C_3L_3R_3s^2+L_3s+R_3}{C_3L_3s^2+1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1C_3L_1L_3R_1R_3g_ms^4 + C_1L_1L_3R_1g_ms^3 + L_3R_1g_ms + R_1R_3g_m + s^2(C_1L_1R_1R_3g_m + C_3L_3R_1R_3g_m)}{C_1C_3L_3R_1s^3 + C_1R_1s + R_1g_m + s^4(C_1C_3L_1L_3R_1g_m + C_1C_3L_1L_3) + s^2(C_1L_1R_1g_m + C_1L_1 + C_3L_3R_1g_m + C_3L_3) + 1}$$

9.90 X-INVALID-ORDER-90 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1},\ \infty,\ \frac{R_3(C_3L_3s^2+1)}{C_3L_3s^2+C_3R_3s+1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1C_3L_1L_3R_1R_3g_ms^4 + R_1R_3g_m + s^2(C_1L_1R_1R_3g_m + C_3L_3R_1R_3g_m)}{R_1g_m + s^4(C_1C_3L_1L_3R_1g_m + C_1C_3L_1L_3) + s^3(C_1C_3L_1R_1R_3g_m + C_1C_3L_1R_3 + C_1C_3L_3R_1) + s^2(C_1C_3R_1R_3 + C_1L_1R_1g_m + C_1L_1 + C_3L_3R_1g_m + C_3L_3) + s(C_1R_1 + C_3R_1R_3g_m + C_3R_3) + 1}$$

10 X-INVALID-WZ

10.1 X-INVALID-WZ-1 $Z(s) = \left(\frac{L_1s}{C_1L_1s^2+1},\ \infty,\ L_3s + \frac{1}{C_3s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3L_1L_3g_ms^2 + L_1g_m}{C_1C_3L_1s^2 + C_3L_1g_ms + C_3}$$

Parameters:

Q: $\frac{\sqrt{C_1}}{\sqrt{L_1g_m}}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{g_m}{C_1}$
K-LP: $\frac{L_1g_m}{C_3}$
K-HP: $\frac{L_3g_m}{C_1}$
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$

10.2 X-INVALID-WZ-2 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2+1},\ \infty,\ L_3 s + R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3 L_1 L_3 g_m s^2 + C_3 L_1 R_3 g_m s + L_1 g_m}{C_1 C_3 L_1 s^2 + C_3 L_1 g_m s + C_3}$$

Parameters:

Q: $\frac{\sqrt{C_1}}{\sqrt{L_1} g_m}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{g_m}{C_1}$
K-LP: $\frac{L_1 g_m}{C_3}$
K-HP: $\frac{L_3 g_m}{C_1}$
K-BP: R_3
Qz: None
Wz: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$

10.3 X-INVALID-WZ-3 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2+L_1 s+R_1},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3 L_1 L_3 R_1 g_m s^2 + L_1 R_1 g_m}{C_1 C_3 L_1 R_1 s^2 + C_3 R_1 + s\left(C_3 L_1 R_1 g_m + C_3 L_1\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1} R_1}{\sqrt{L_1} R_1 g_m + \sqrt{L_1}}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{\sqrt{L_1} R_1 g_m + \sqrt{L_1}}{C_1 \sqrt{L_1} R_1}$
K-LP: $\frac{L_1 g_m}{C_3}$
K-HP: $\frac{L_3 g_m}{C_1}$
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$

10.4 X-INVALID-WZ-4 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2+L_1 s+R_1},\ \infty,\ L_3 s + R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3 L_1 L_3 R_1 g_m s^2 + C_3 L_1 R_1 R_3 g_m s + L_1 R_1 g_m}{C_1 C_3 L_1 R_1 s^2 + C_3 R_1 + s\left(C_3 L_1 R_1 g_m + C_3 L_1\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1} R_1}{\sqrt{L_1} R_1 g_m + \sqrt{L_1}}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{\sqrt{L_1} R_1 g_m + \sqrt{L_1}}{C_1 \sqrt{L_1} R_1}$
K-LP: $\frac{L_1 g_m}{C_3}$
K-HP: $\frac{L_3 g_m}{C_1}$
K-BP: $\frac{R_1 R_3 g_m}{R_1 g_m + 1}$
Qz: None
Wz: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$

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